

objective?

	Course 1	Course 2	Course 3	Course 4
Prof 1	60	40	60	70
Prof 2	20	60	50	70
Prof 3	20	30	40	60
TA	30	10	20	40

b) Solve the following LPP using Simplex algorithm.

$$\text{Max } Z = -2x + 3y$$

subject to

$$x \leq 5$$

$$2x - 3y \leq 6$$

$$x, y \geq 0$$

Q9. a) Find the initial basic feasible solution of the following transportation problem using Vogel's Approximation method. [7+8=15]

	I	II	III	IV	Supply
A	21	16	25	13	11
B	17	18	14	23	13
C	32	27	18	41	19
Demand	6	10	12	15	

b) Three food products are available at costs of Rs. 10, Rs. 36 and Rs. 24 per unit, respectively. They contain 1,000, 4,000 and 2,000 calories per unit, respectively. It is required to find the minimum-cost diet containing at least 20,000 calories and 3,000 units of protein. Formulate and solve the given problem as an LP problem. Write the dual and using the final simplex table of primal problem, write the solution to the dual problem.

(Please write your Exam Roll No.)

Exam Roll No.

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: BS-202

Subject: Probability, Statistics & Linear Programming

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No 1 which is compulsory. Select one question from each unit. Assume missing data if any.

[Scientific Calculators are allowed.]

Q1. Attempt all Questions:-

3

a) A batch of 25 injection-molded parts contains 5 that have suffered excessive shrinkage. If two parts are selected at random and without replacement, what is the probability that the second part selected is one with excessive shrinkage?

b) State the Central Limit Theorem.

2

c) Find the moment-generating function of a random variable X following an exponential distribution.

3

d) Explain the general procedure for Hypothesis Testing.

2

e) Solve the following LPP graphically.

5

$$\text{Min } Z = -3x + 2y$$

subject to

$$x + y \leq 5$$

$$4 \geq x \geq 0, 6 \geq y \geq 1$$

UNIT I

Q2. a) Let the random variable X denote the current measured in a thin copper wire in milliamperes. Assume that the range of X is $[0, 20 \text{ mA}]$, and assume that the probability density function of X is $f(x) = 0.05, 0 \leq x \leq 20$. What is the probability that a measurement of current is between 5 and 10 milliamperes? Also, determine the mean and standard deviation of X . [7+8=15]

b) Customers are used to evaluate preliminary product designs. In the past, 95% of highly successful products received good reviews, 60% of moderately successful products received good reviews, and 10% of poor products received good reviews. In addition, 40% of products have been highly successful, 35% have been moderately successful, and 25% have been poor products.

i. If a new design attains a good review, then what is the probability that it will be a highly successful product?

P-4/4

P-4/4

P.T.O.

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: CIC-206

Subject: Theory of Computation

Time: 3 Hours

Maximum Marks: 75

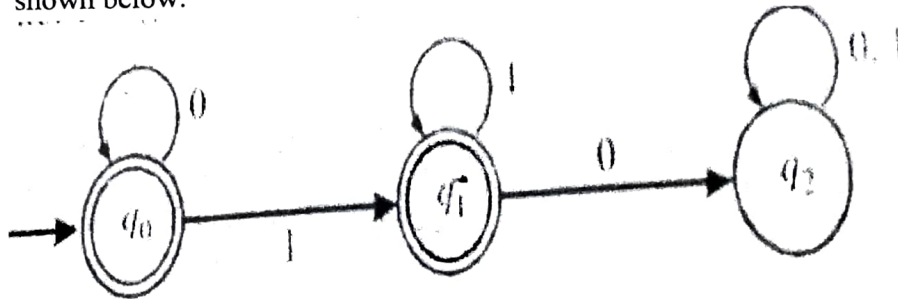
Note: Attempt five questions in all including Q.No.1 which is compulsory. Assume missing data, if any.

(2.5x10=25)

Q1 Answer the following questions:

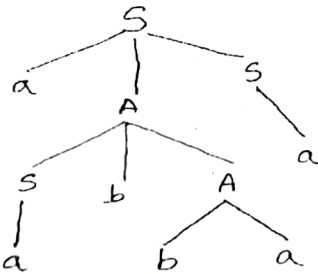
- Write any five Identity Rules for Regular Expressions?
- What is the difference between Deterministic Finite Automata (DFA) and Non-Deterministic Finite Automata (NDFA)?
- What is the disadvantages of unambiguous parse tree. Give an example.
- Define the term context free grammar with an example?
- Define instantaneous description of a PDA
- Construct the CFG for generating the language $L = \{a^n b^n / n \geq 1\}$.
- For the grammar $S \rightarrow A1B, A \rightarrow 0A \mid \epsilon, B \rightarrow 0B \mid 1B \mid \epsilon$, give leftmost and rightmost derivations for the string 00101.
- Prove graph coloring problem is NP complete.
- Describe the Set of all Strings of 0's and 1's ending in 00 by Regular Expression.
- Construct a Regular Grammar G generating the regular expression $0^*(0+1)^*$?

Q2 a) Construct a Regular Expression for the FA (Using Method of Eliminate States) the shown below: (6)



b) Explain Pumping lemma for Context Free Languages. (6.5)

Q3 a) Find the left most and right most derivation corresponding to the tree. (6)



b) Construct a equivalent grammar G in CNF for the grammar G1 where $G1 = (\{S, A, B\}, \{a, b\}, \{S \rightarrow bA/aB, A \rightarrow bAA/aS/a, B \rightarrow aBB/bS/b\}, S)$ (6.5)

P.T.O.

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: ETCS-206

Subject: Theory of Computation

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Assume missing data, if any.

- Q1 Differentiate between the following: [5×5=25]
- a) Regular Language and Context Free language
 - b) Recursive and Recursive Enumerable language
 - c) Halting Problem & Un-decidable problem of Turing machine
 - d) Classify & Differentiate between different types of FA with output.
 - e) Dead state and In-accessible state
- Q2
- a) Explain closure properties of regular sets. Provide at least two closure properties of regular sets and justify their significance in formal language theory. [4]
 - b) Convert the following regular expression to DFA.
 $ab(a+b)^*ab$ [4]
 - c) Explain the significance of the stack in a Pushdown Automaton with example. [4.5]
- Q3
- a) How the ambiguity of a grammar can be removed? Explain with suitable example. [6]
 - b) How to Convert any CFG to GNF? Explain with suitable example. [6.5]
- Q4
- a) What are Normal Forms in grammar? Explain its importance with example. [4]
 - b) Explain two types of acceptance in pushdown automata for $a^n b^n$. [4.5]
 - c) Define Pushdown Automata and explain its components. [4]
- Q5 Write a short note on following
- a) Kleen's closure & Positive Closure [4]
 - b) Undecidability [4.5]
 - c) SPACE and PSPACE [4]
- Q6
- a) Write the steps and convert the regular grammar into Finite Automata for
 $S \rightarrow 0A/1B0/1$ $A \rightarrow 0S/1B/1$ $B \rightarrow 0A/S$ [6.5]
 - b) Construct a TM for $a^n b^n$. [6]
- Q7
- a) Design a Turing machine for string of palindrome (Same machine for even and odd length). [7.5]
 - b) Write a short note on Computability and Intractability. [5]

P.T.O.

P 1/2
ETCS-206

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: EEC-208 Subject: Circuits and Systems

Time: 3 Hours Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Attempt all questions:

- Find the Fourier transform for $x(t) = \sin(\omega t) u(t)$ (3)
- List the properties of ROC. (3)
- Determine the impulse response of series and parallel RL circuit. (3)
- Explain the AC circuit containing the inductor only. (3)
- Explain short circuit and open circuit impedance in terms of ABCD parameters. (3)

UNIT-I

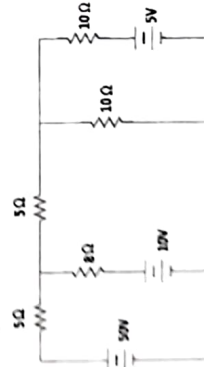
Q2 Examine the following system: Linear/Nonlinear, Causal/Non-Causal, Time Variant/Time-Invariant, and Static/Dynamic with examples. (15)

Q3 Find the Laplace transform for following signals if $x(t) = e^{-at} u(t)$. (5×3=15)

- $y(t) = x(t + 2)$
- $y(t) = tx(t)$
- $y(t) = dx(t)/dt$

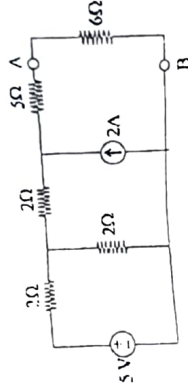
UNIT-IIQ4 a) Derive the Transient Response in series R-L-C circuit with sinusoidal excitation for second order circuit. (7)
b) A step voltage of 20 V is applied at $t = 0$ in an R-L-C series circuit where $R = 50\Omega$, $L = 1H$ and $C = 1F$. Calculate $i(t)$ with and without Laplace Transform. (8)Q5 a) Find the step response (by applying a constant voltage source at the close of a switch) of a Parallel RLC circuit. (8)
b) Determine the step response of series and parallel RC circuits. (7)**UNIT-III**

Q6 For the Circuit shown below, determine the current in each branch using Nodal and Loop analysis. (15)



P.T.O.

[-2-]

Q7 Determine the value of current in 6Ω resistor using Thevenin's theorem and Norton's theorem. (15)**UNIT-IV**Q8 a) Explain the term 'tie-set' and 'tie-set matrix' of a network with an illustrative example. (5)
b) Derive and prove the following Matrix Equation with an appropriate example $I_b = B^T I_L$ (5)Where, B is the Tie-set matrix, I_b and I_L are the branch current respectively (10)Q9 a) Derive the transmission parameters of Two-port networks. Also, determine the condition of Reciprocity and symmetry for ABCD parameter. (8)
b) Explain the parallel and cascade connection of Two-port networks. (7)

P-1/2
EEC-208P-2/2
EEC-208

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: CIC-210 Subject: Database Management System
 Time: 3 Hours Maximum Marks: 75

Note: Attempt questions in all including Q.No.1 which is compulsory. Select one question from each Unit.

- Q1 a) Give an example of mapping of generalization or specialization into relation schemas. Summarize the steps involved in converting the ER constructs to relational schema. (3)
- b) What is a Weak Entity type? Explain the role of partial key in design of weak entity type. (3)
- c) Consider a relational schema with the following tables:
 Employee (EmpID, EmpName, DeptID)
 Department (DeptID, DeptName)
 Write a relational calculus expression to retrieve the names of all employees who work in the "Sales" department. (3)
- d) Why is the domain key normal form (DKNF) known as ultimate normal form? (3)
- e) Compare and contrast heap files and sorted files in terms of storage efficiency and query performance. When would you choose to use a heap file over a sorted file, and vice versa? (3)

UNIT-I

- Q2 a) Design an ER — Diagram for a UNIVERSITY database schema taking into account at least five entities and indicate all keys and cardinality constraints and assumptions that are made. (7)
- b) Define the following with examples:
 (i) Super key (ii) Candidate key
 (iii) Primary key (iv) Foreign key (5)
- c) What do you mean by constraints? Differentiate between table and column constraint. Explain with an example. (3)

- Q3 a) Consider the following tables: (8)

works (Pname, Cname, Salary)
 lives (Pname, Street, City)
 located-In (Cname, City)

write the following queries in SQL:

- i) List the names of the people who work for the company 'Wipro' along with the cities they live in.
- ii) Find the names of the persons who do not work for 'Infosys'.
- iii) Find the people whose salaries are more than that of all of the 'oracle' employees.
- iv) Find the persons who works and lives in the same city.
- b) Differentiate between DDL, DML, and DCL in SQL. Provide examples for each category and explain their roles in database management. (4)
- c) Explain the various inner join operations in relational algebra with examples. (3)

P.T.O.

P-13
 CIC-210

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Subject: Programming in Java

Paper Code: CIC-212

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Answer the following questions: (5x3=15)

- Discuss any five differences between Interface and Abstract Class.
- Explain the concept of wrapper classes in Java and discuss their significance?
- What is remote method invocation? Explain briefly.
- How to create Inner & Outer Class? How it is helpful for achieving Security.
- What is the thread's priority? Define the values for different priority in java.

UNIT-I

- Q2 a) What is JVM? Explain JVM architecture in detail. (8)
- b) What is Sandbox security model? How does it help in providing security to resources? Discuss in detail. (7)
- Q3 a) Explain the security promises of the Java Virtual Machine (JVM) and how it achieves them. (5)

- b) Write a short note on the following: (10)
- Various Instructions sets used in Java.
 - Garbage Collections
 - Class File Format

UNIT-II

- Q4 a) Write a Java program demonstrating the implementation of an interface. Define an interface named Shape with methods calculate Area() and calculate Perimeter(). Implement this interface with classes Circle and Rectangle. Test the implementation by creating objects of these classes and invoking the interface methods to calculate their respective areas and perimeters. (7)
- b) What is exception handling? Create customized exceptions and make use of all the 5 exception keywords used in exception handling. (8)
- Q5. a) Explain the concept of Applet programming in Java and describe the life cycle of an applet. Provide a Java Applet program illustrating the key stages of an applet's life cycle. (8)
- b) What is the difference between String Buffer and String Class? Show the use of two methods of String Buffer and String class with the help of example. (7)

P.T.O.

[-2-] UNIT-III

- Q6. a) What is thread? Explain the life cycle of a thread with the help of a diagram. What are the various ways to create the thread? Explain to anyone with the help of example. (8)
- b) What is thread synchronization? When do we use it? Write a program to illustrate the concept of Thread Synchronization method (7)
- Q7. a) Why is event used in Java? What is the difference between adapter class and listener interface? Explain with the help of an example. (8)
- b) Discuss various types of layouts used in AWT. Write a program to show the use of Border Layout and Grid Layout with various components. (7)

UNIT-IV

- Q8. a) Explain different types of JDBC drivers. Write the steps to connect a Java application to a database? Explain with the help of example. (10)
- b) Write a program to sort the content of a given text file. (5)
- Q9. a) Write a program to implement the Client -Server Application. (7)
- b) Write the short note on:
(i)What is JNI and how it works?
(ii)What is object serialization vs deserialization?
(iii)What are Java collections? Give some methods of Collection interface. (8)

P-1/2
CIC-212

P-2/2

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] JUNE 2024

Paper Code: ETEC-212

Subject: Communication Systems

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q.no.1 which is compulsory.

- Q1 Attempt all questions. (5x5=25)
- (a) What is meant by Ergodic process
 - (b) Define a random variable. Specify the sample space and the random variable for a coin tossing experiment
 - (c) State the reasons for higher noise in mixers
 - (d) What are the types of analog modulation
 - (e) What are the advantages of VSB-AM
- Q2 (a) What is joint probability and conditional probability. Explain (6)
(b) Mention any two properties of Gaussian Process (6.5)
- Q3 (a) State any four properties of power spectral density (6.5)
(b) Explain poisson process and weiner process (6)
- Q4 (a) Explain analog modulation with suitable diagram and why we need modulation index (6)
(b) Explain the DSB-SC modulation and demodulation with suitable diagram (6.5)
- Q5 (a) Explain coherent detection and their drawbacks (6)
(b) A broadcast radio transmitter radiates 5 kW power when the modulation percentage is 60 percent. How much is the carrier power (6.5)
- Q6 (a) A carrier frequency is frequency modulated with a sinusoidal signal of 2 KHz resulting in a maximum frequency deviation of 5 KHz. Find the approximate band width of the modulated signal (6)
(b) Distinguish between narrow band FM and wide band FM (6.5)
- Q7 (a) What are the two methods of producing FM waves (6)
(b) Explain ratio detector with diagram (6.5)
- Q8 (a) How does pre-emphasis and de-emphasis process provide overall SNR improvement in FM systems (6)
(b) Write short note on (6.5)
(i) capture effect
(ii) threshold effect
- Q9 (a) What is external noise? List the external sources of noise (6)
(b) How does pre-emphasis and de-emphasis process provide overall SNR improvement in FM systems (6.5)

ETEC-212